

README Dataset for: Boosting hydrogen storage and release in MOF-5 / graphite hybrids via in situ synthesis

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This replication dataset gathers experimental measurements evaluating the performance of pristine MOF-5 metal-organic frameworks and their exfoliated graphite (EG) hybrids for hydrogen (H₂) storage and delivery. This README document provides comprehensive documentation for all raw and processed experimental data characterizing the synthesized materials (including FTIR, TGA/DTG, pore size distributions, and N₂, Ar, and H₂ adsorption isotherms across cryogenic to room temperature ranges). Within this document, you will find a detailed data dictionary, a complete inventory and description of the uploaded files, as well as the sample naming conventions and nomenclature used for all pristine MOF-5, EG, and MOF-5@EG composite materials evaluated throughout this study.

File Inventory & Description:

X-Ray Diffraction (XRD) Patterns

This section contains experimental X-ray diffraction (XRD) patterns for pristine MOF-5, exfoliated graphite (EG), and their carbon-based hybrids. The datasets report diffraction intensity versus scattering angle (2θ) to verify crystalline structure, phase purity, and framework integrity after carbon nanostructure incorporation.

XRD-- MOF-5.tab: X-ray diffraction (XRD) pattern of MOF-5

XRD -- MOF-5 + 1%EG.tab: X-ray diffraction (XRD) pattern for MOF-5 + 1% exfoliated graphite

XRD -- MOF-5 + 5%EG.tab: X-ray diffraction (XRD) pattern for MOF-5 + 5% exfoliated graphite

XRD -- MOF-5 + 10%EG.tab: X-ray diffraction (XRD) pattern for MOF-5 + 10% exfoliated graphite

XRD – EG.tab: X-ray diffraction (XRD) pattern of EG

FTIR Spectra Data

This section contains experimental FTIR spectra for pristine MOF-5 and its exfoliated graphite (EG) hybrids. The datasets report transmittance versus wavenumber (cm⁻¹) to evaluate how EG % (1 wt.%, 5 wt.% and 10 wt.%) influence the functional groups and chemical bonds of the materials.

FTIR.tab: FTIR spectra of pure MOF-5 and EG/MOF-5 composites (1 wt.%, 5 wt.% and 10 wt.%)

Thermogravimetric Analysis (TGA & DTG)

This section contains TGA and DTG curves for pristine MOF-5, exfoliated graphite (EG), and their carbon-based hybrids. The datasets report mass loss percentage and rate of mass loss versus temperature to evaluate how EG % (1 wt.%, 5 wt.% and 10 wt.%) influence thermal stability and decomposition behavior.

TGA-- MOF-5.tab: Thermogravimetric analysis (TGA) profile for MOF-5

TGA -- MOF-5 + 1%EG.tab: Thermogravimetric analysis (TGA) profile for MOF-5 + 1% exfoliated graphite

TGA -- MOF-5 + 5%EG.tab: Thermogravimetric analysis (TGA) profile for MOF-5 + 5% exfoliated graphite

TGA -- MOF-5 + 10%EG.tab: Thermogravimetric analysis (TGA) profile for MOF-5 + 10% exfoliated graphite

TGA -- EG.tab: Thermogravimetric analysis (TGA) profile of EG

DTG-- MOF-5.tab: Derivative thermogravimetric (DTG) profile for MOF-5

DTG-- EG.tab: Derivative thermogravimetric (DTG) profile of EG

DTG-- MOF-5 + 1%EG.tab: Derivative thermogravimetric (DTG) profile for MOF-5 + 1% exfoliated graphite

DTG-- MOF-5 + 5%EG.tab: Derivative thermogravimetric (DTG) profile for MOF-5 + 5% exfoliated graphite

DTG-- MOF-5 + 10%EG.tab: Derivative thermogravimetric (DTG) profile for MOF-5 + 10% exfoliated graphite

Argon Adsorption/Desorption Isotherms (87 K)

This section contains experimental Argon (Ar) adsorption and desorption isotherms at 87 K for pristine MOF-5 and its exfoliated graphite (EG) hybrids. The datasets report gas uptake versus relative pressure (p/p) to evaluate how EG % (1 wt.%, 5 wt.% and 10 wt.%) influence cryo-adsorptive performance.

AR-- MOF-5.tab: Ar adsorption-desorption isotherms at 87 K for MOF-5

AR-- MOF-5 + 1%EG.tab: Argon adsorption-Desorption Isotherms at 87K for MOF-5 + 1% exfoliated graphite

AR-- MOF-5 + 5%EG.tab: Argon adsorption-Desorption Isotherms at 87K for MOF-5 + 5% exfoliated graphite

AR-- MOF-5 + 10%EG.tab: Argon adsorption-Desorption Isotherms at 87K for MOF-5 + 10% exfoliated graphite

Nitrogen Adsorption/Desorption Isotherms (77 K)

This section contains experimental Nitrogen (N₂) adsorption and desorption isotherms at 77 K for pristine MOF-5 and its exfoliated graphite (EG) hybrids. The

datasets report gas uptake versus relative pressure (p/p) to evaluate how EG % (1 wt.%, 5 wt.% and 10 wt.%) influence cryo-adsorptive performance and textural properties.

N2-- MOF-5.tab: Nitrogen adsorption/desorption isotherms at 77 K for MOF-5

N2-- MOF-5 + 1%EG.tab: Nitrogen adsorption/desorption isotherms at 77 K for MOF-5 + 1% exfoliated graphite

N2-- MOF-5 + 5%EG.tab: Nitrogen adsorption/desorption isotherms at 77 K for MOF-5 + 5% exfoliated graphite

N2-- MOF-5 + 10%EG.tab: Nitrogen adsorption/desorption isotherms at 77 K for MOF-5 + 10% exfoliated graphite

Water Adsorption/Desorption Isotherms (77 K)

This section contains experimental water (H₂O) adsorption and desorption isotherms at 298 K for pristine MOF-5 and its exfoliated graphite (EG) hybrids. The datasets report gas uptake versus absolute pressure to evaluate how EG % (1 wt.%, 5 wt.% and 10 wt.%) influence the water adsorption performance, hydrophilicity, and textural properties.

H2O--MOF-5.tab: H₂O adsorption isotherms at 298 K of MOF-5

H2O -- MOF-5+ 1%EG.tab: H₂O adsorption isotherms at 298 K for MOF-5 + 1% exfoliated graphite

H2O -- MOF-5+ 5%EG.tab: H₂O adsorption isotherms at 298 K for MOF-5 + 5% exfoliated graphite

H2O -- MOF-5+ 10%EG.tab: H₂O adsorption isotherms at 298 K for MOF-5 + 10% exfoliated graphite

Low-Pressure Hydrogen Adsorption/Desorption Isotherms (77 K)

This section contains experimental low-pressure Hydrogen (H₂ up to 1 bar) adsorption and desorption isotherms at 77 K for pristine MOF-5 and its exfoliated graphite (EG) hybrids. The datasets report gas uptake versus absolute pressure to evaluate how GO % (1 wt.%, 5 wt.% and 10 wt.%) influence cryo-adsorptive performance.

H2-- MOF-5.tab: H₂ adsorption-desorption isotherms at 77K and 1 bar for MOF-5

H2-- MOF-5+ 1%EG.tab: H₂ adsorption-desorption isotherms at 77K and 1 bar for MOF-5 + 1% exfoliated graphite

H2-- MOF-5+ 5%EG.tab: H₂ adsorption-desorption isotherms at 77K and 1 bar for MOF-5 + 5% exfoliated graphite

H2-- MOF-5+ 10%EG.tab: H₂ adsorption-desorption isotherms at 77K and 1 bar for MOF-5 + 10% exfoliated graphite

Excess Hydrogen Adsorption Isotherms (77, 160, and 273 K)

This section contains experimental excess hydrogen (H₂) adsorption isotherms for pristine MOF-5 and its exfoliated graphite (EG) hybrids measured at 77 K, 160 K, and 273 K (up to 130 bar). The datasets report gas uptake versus absolute pressure to evaluate how EG % (1 wt.%, 5 wt.% and 10 wt.%) influence hydrogen storage performance across cryogenic to room temperature ranges.

EXCESS H2 77K-- MOF-5.tab: Excess H₂ adsorption isotherms at 77 K (up to 130 bar) for MOF-5

EXCESS H2 77K-- MOF-5+ 1%EG.tab: Excess H₂ adsorption isotherms at 77 K (up to 130 bar) for MOF-5 + 1% exfoliated graphite

EXCESS H2 77K-- MOF-5+ 5%EG.tab: Excess H₂ adsorption isotherms at 77 K (up to 130 bar) for MOF-5 + 5% exfoliated graphite

EXCESS H2 77K-- MOF-5+ 10%EG.tab: Excess H₂ adsorption isotherms at 77 K (up to 130 bar) for MOF-5 + 10% exfoliated graphite

EXCESS H2 160K-- MOF-5.tab: Excess H₂ adsorption isotherms at 160 K (up to 130 bar) for MOF-5

EXCESS H2 160K-- MOF-5+ 1%EG.tab: Excess H₂ adsorption isotherms at 160 K (up to 130 bar) for MOF-5 + 1% exfoliated graphite

EXCESS H2 160K-- MOF-5+ 5%EG.tab: Excess H₂ adsorption isotherms at 160 K (up to 130 bar) for MOF-5 + 5% exfoliated graphite

EXCESS H2 160K-- MOF-5+ 10%EG.tab: Excess H₂ adsorption isotherms at 160 K (up to 130 bar) for MOF-5 + 10% exfoliated graphite

EXCESS H2 273K-- MOF-5.tab: Excess H₂ adsorption at 273 K (up to 130 bar) for MOF-5

EXCESS H2 273K-- MOF-5+ 1%EG.tab: Excess H₂ adsorption at 273 K (up to 130 bar) for MOF-5 + 1% exfoliated graphite

EXCESS H2 273K-- MOF-5+ 5%EG.tab: Excess H₂ adsorption at 273 K (up to 130 bar) for MOF-5 + 5% exfoliated graphite

EXCESS H2 273K-- MOF-5+ 10%EG.tab: Excess H₂ adsorption at 273 K (up to 130 bar) for MOF-5 + 10% exfoliated graphite

Total Hydrogen Adsorption Isotherms (77 K)

This section contains total hydrogen (H₂) adsorption isotherms for pristine MOF-5 and its exfoliated graphite (EG) hybrids measured at 77 K (up to 130 bar). The datasets report gas uptake versus absolute pressure to evaluate how EG % (1 wt.%, 5 wt.% and 10 wt.%) influence hydrogen storage performance.

TOTAL H2 77K-- MOF-5.tab: Total H₂ adsorption at 77 K (up to 130 bar) for MOF-5

TOTAL H2 77K-- MOF-5+ 1%EG.tab: Total H2 adsorption at 77 K (up to 130 bar) for MOF-5 + 1% exfoliated graphite

TOTAL H2 77K-- MOF-5+ 5%EG.tab: Total H2 adsorption at 77 K (up to 130 bar) for MOF-5 + 5% exfoliated graphite

TOTAL H2 77K-- MOF-5+ 10%EG.tab: Total H2 adsorption at 77 K (up to 130 bar) for MOF-5 + 10% exfoliated graphite

Pore Size Distribution Data

This section contains pore size distribution data for pristine MIL-101(Cr) and its graphene oxide (GO) hybrids. The datasets report differential pore volume versus pore width how GO % (1 wt.%, 5 wt.% and 10 wt.%) influence the porous network framework.

PSD-- MOF-5.tab: Pore size distribution (PSD) for MOF-5

PSD-- MOF-5+ 1%EG.tab: Pore size distribution (PSD) for MOF-5 + 1% exfoliated graphite

PSD-- MOF-5+ 5%EG.tab: Pore size distribution (PSD) for MOF-5 + 5% exfoliated graphite

PSD-- MOF-5+ 10%EG.tab: Pore size distribution (PSD) for MOF-5 + 10% exfoliated graphite

Optical microscope images

This section contains optical microscopy data and images for pristine MOF-5 and its exfoliated graphite (EG) hybrids. These datasets document the visual morphology and structural changes of the framework across different EG loadings (1 wt.%, 5 wt.%, and 10 wt.%).

MOF-5.jpg: Optical microscope image for MOF-5

EG.jpg: Optical microscope image for EG

MOF-5+ 1%EG.jpg.tab: Optical microscope image for MOF-5 + 1% exfoliated graphite

MOF-5+ 5%EG.jpg: Optical microscope image for MOF-5 + 5% exfoliated graphite

MOF-5+ 10%EG.jpg: Optical microscope image for MOF-5 + 10% exfoliated graphite

Sample naming conventions:

MOF-5: Pristine MOF-5 synthesized via in situ route.

EG: Exfoliated graphite

MOF-5 + 1%EG: Composite obtained via in situ synthesis of MOF-5 and 1 wt.% EG.

MOF-5 + 5%EG: Composite obtained via in situ synthesis of MOF-5 and 5 wt.% EG.

MOF-5 + 10%EG: Composite obtained via in situ synthesis of MOF-5 and 10 wt.% EG.